

Math 163, Winter 2026/Spring 2026
SCRIPT 13: Uniform Continuity and Integration

We will now consider a notion of continuity that is stronger than ordinary continuity.

Definition 13.1. Let $f: A \rightarrow \mathbb{R}$ be a function. We say that f is *uniformly continuous* if for all $\epsilon > 0$, there exists a $\delta > 0$ such that for all $x, y \in A$,

$$\text{if } |x - y| < \delta, \quad \text{then } |f(x) - f(y)| < \epsilon.$$

Theorem 13.2. *If f is uniformly continuous, then f is continuous.*

Proof of Theorem 13.2. Let f be uniformly continuous. We want to show f is continuous.

Since f is uniformly continuous, for all $\epsilon > 0$, there exists $\delta > 0$ such that for all $x, y \in A$, if $|x - y| < \delta$ then $|f(x) - f(y)| < \epsilon$.

Let $x_0 \in A$ and let $\epsilon > 0$. Since f is uniformly continuous, there exists $\delta > 0$ such that for all $x, y \in A$, if $|x - y| < \delta$ then $|f(x) - f(y)| < \epsilon$. In particular, for all $y \in A$, if $|y - x_0| < \delta$ then $|f(y) - f(x_0)| < \epsilon$. Hence f is continuous at x_0 .

Since $x_0 \in A$ was arbitrary, f is continuous on A . □

Exercise 13.3. Determine with proof whether each function f is uniformly continuous on the given interval A .

- (a) $f(x) = x^2$ on $A = \mathbb{R}$.
- (b) $f(x) = x^2$ on $A = (-2, 2)$.
- (c) $f(x) = \frac{1}{x}$ on $A = (0, +\infty)$.
- (d) $f(x) = \frac{1}{x}$ on $A = [1, +\infty)$.
- (e) $f(x) = \sqrt{x}$ on $A = [1, +\infty)$.

Exercise (a.) We claim that f is not uniformly continuous. Let $\epsilon = 1$ and suppose for sake of contradiction that there exists $\delta > 0$ such that for all $j, k \in \mathbb{R}$, if $|j - k| < \delta$ then $|f(j) - f(k)| < \epsilon$, that is, $|j^2 - k^2| < 1$.

By the Archimedean property, there exists $n \in \mathbb{N}$ such that $0 < \frac{1}{n} < \delta$. We note that $|(n + \frac{1}{n}) - n| = \frac{1}{n} < \delta$, hence by assumption,

$$\left| \left(n + \frac{1}{n} \right)^2 - n^2 \right| = 2 + \frac{1}{n^2} < 1.$$

We can therefore write $2 < 1 - \frac{1}{n^2}$. Since $n \in \mathbb{N}$, we have that $\frac{1}{n^2} > 0$, hence $1 - \frac{1}{n^2} < 1$, and therefore $2 < 1$, which is a contradiction. This completes the proof. $\square$

Exercise (b.) We claim that f is uniformly continuous on $(-2, 2)$. Let $x, y \in (-2, 2)$. We note that

$$|f(x) - f(y)| = |x^2 - y^2| = |x + y||x - y|.$$

Since $x, y \in (-2, 2)$, we have that $|x + y| \leq |x| + |y| < 4$. Therefore,

$$|f(x) - f(y)| = |x + y||x - y| < 4|x - y|.$$

Let $\delta = \frac{\epsilon}{4}$. Then, we have that $|f(x) - f(y)| < 4|x - y| < 4\delta = \epsilon$. This completes the proof. $\square$

Exercise (c.) We claim that f is not uniformly continuous. Let $\epsilon = 1$ and suppose for sake of contradiction that there exists $\delta > 0$ such that for all $j, k \in (0, +\infty)$, if $|j - k| < \delta$ then $|f(j) - f(k)| < \epsilon$, that is, $|\frac{1}{j} - \frac{1}{k}| < 1$. Without loss of generality, suppose $\delta < 1$.

Let $j = \frac{\delta}{4}$ and $k = \frac{3\delta}{4}$. Then, we have that $|j - k| = \frac{\delta}{2} < \delta$. Furthermore, we compute

$$\left| \frac{1}{j} - \frac{1}{k} \right| = \left| \frac{4}{\delta} - \frac{4}{3\delta} \right| = \frac{4}{\delta} \cdot \frac{2}{3} = \frac{8}{3\delta}.$$

Since $\epsilon = 1$, by assumption we have that $\frac{8}{3\delta} < 1$, which implies $\delta > \frac{8}{3}$. This contradicts the assumption that $\delta < 1$. This completes the proof. $\square$

Exercise (d.) We claim that f is uniformly continuous on $[1, +\infty)$. Let $x, y \in [1, +\infty)$. We note that

$$\left| \frac{1}{x} - \frac{1}{y} \right| = \left| \frac{x - y}{xy} \right| = \frac{|x - y|}{xy} \leq |x - y|,$$

where the last inequality follows since $x, y \geq 1$ implies $xy \geq 1$. Let $\delta = \epsilon$. Then, we have that $|\frac{1}{x} - \frac{1}{y}| \leq |x - y| < \delta = \epsilon$. This completes the proof. $\square$

Exercise (e.) We claim that f is uniformly continuous on $[1, +\infty)$. Let $x, y \in [1, +\infty)$. We note that

$$(\sqrt{x} - \sqrt{y})(\sqrt{x} + \sqrt{y}) = x - y,$$

hence $|\sqrt{x} - \sqrt{y}| = \frac{|x - y|}{\sqrt{x} + \sqrt{y}}$. Since $x, y \in [1, +\infty)$, we have that $\sqrt{x} + \sqrt{y} \geq 2$, and therefore

$$|\sqrt{x} - \sqrt{y}| = \frac{|x - y|}{\sqrt{x} + \sqrt{y}} \leq \frac{|x - y|}{2}.$$

Let $\delta = 2\epsilon$. Then, we have that $|\sqrt{x} - \sqrt{y}| \leq \frac{|x - y|}{2} < \frac{\delta}{2} = \epsilon$. This completes the proof. $\square$

Exercise 13.4. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be defined by $f(x) = x^n$ for $n \in \mathbb{N}$. Show that f is uniformly continuous if, and only if, $n = 1$.

Exercise 13.4. ($\Leftarrow$) Suppose $n = 1$, so $f(x) = x$. Let $\epsilon > 0$ and set $\delta = \epsilon$. Then for all $x, y \in \mathbb{R}$, if $|x - y| < \delta$ then $|f(x) - f(y)| = |x - y| < \delta = \epsilon$. Hence f is uniformly continuous.

($\Rightarrow$) We prove the contrapositive: if $n > 1$, then f is not uniformly continuous. Suppose $n \geq 2$. Let $\epsilon = 1$ and suppose for contradiction that there exists $\delta > 0$ such that for all $x, y \in \mathbb{R}$, if $|x - y| < \delta$ then $|f(x) - f(y)| < \epsilon$.

By the Archimedean property, there exists $N \in \mathbb{N}$ with $\frac{1}{N} < \delta$. Set $x = N + \frac{1}{N}$ and $y = N$. Then $|x - y| = \frac{1}{N} < \delta$, so by assumption $|f(x) - f(y)| < 1$. Then, by the work done in **Exercise 12.7**,

$$\left(N + \frac{1}{N}\right)^n - N^n = \frac{1}{N} \left(\sum_{i=0}^{n-1} \left(N + \frac{1}{N}\right)^{n-1-i} N^i \right).$$

Since $N + \frac{1}{N} > N > 0$, each term in the sum satisfies

$$\left(N + \frac{1}{N}\right)^{n-1-i} N^i > N^{n-1-i} \cdot N^i = N^{n-1},$$

so the sum has n terms each exceeding N^{n-1} , giving

$$\left(N + \frac{1}{N}\right)^n - N^n > \frac{1}{N} \cdot nN^{n-1} = n.$$

Since $n \geq 2$, we get $|f(x) - f(y)| > n \geq 2 > 1 = \epsilon$, a contradiction.

Therefore, $f(x) = x^n$ is uniformly continuous on $\mathbb{R}$ if and only if $n = 1$. □

Challenge: Let $p: \mathbb{R} \rightarrow \mathbb{R}$ be a polynomial with real coefficients. Show that p is uniformly continuous on $\mathbb{R}$ if and only if $\deg(p) \leq 1$.

Exercise 13.5. Let f and g be uniformly continuous on $A \subset \mathbb{R}$. Show that:

- (a) The function $f + g$ is uniformly continuous on A .
- (b) For any constant $c \in \mathbb{R}$, the function $c \cdot f$ is uniformly continuous on A .

Exercise 13.5(a.) We want to show that the function $f + g$ is uniformly continuous on A .

As f is uniformly continuous, we have that $\forall \epsilon > 0$, there exists $\delta_1 > 0$ such that for all $x, y \in A$, if $|x - y| < \delta_1$ then $|f(x) - f(y)| < \frac{\epsilon}{2}$.

Similarly, as g is uniformly continuous, $\forall \epsilon > 0$, there exists $\delta_2 > 0$ such that for all $x, y \in A$, if $|x - y| < \delta_2$ then $|g(x) - g(y)| < \frac{\epsilon}{2}$.

Now, let $\delta = \min(\delta_1, \delta_2)$, and if $|x - y| < \delta$ then $|(f+g)(x) - (f+g)(y)| \leq |f(x) - f(y)| + |g(x) - g(y)| < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon$.

This suffices to complete the proof. $\square$

Exercise 13.5(b.) We want to show that for any constant $c \in \mathbb{R}$, the function $c \cdot f$ is uniformly continuous on A .

As g is uniformly continuous, we have that $\forall \epsilon > 0$, there exists $\delta > 0$ such that if $|x - y| < \delta$ then $|f(x) - f(y)| < \frac{\epsilon}{|c|+1}$. Then, $|c \cdot f(x) - c \cdot f(y)| = |c| |f(x) - f(y)| < |c| \cdot \frac{\epsilon}{|c|+1} < \epsilon$.

This suffices to complete the proof. $\square$

Theorem 13.6. Suppose that $X \subset \mathbb{R}$ is compact and $f: X \rightarrow \mathbb{R}$ is continuous. Then f is uniformly continuous.

Proof. Let $\epsilon > 0$. Since f is continuous, we have that for all $x \in X$, there exists $\delta_x > 0$ such that if $y \in X$ and $|x - y| < \delta_x$ then $|f(x) - f(y)| < \frac{\epsilon}{2}$.

Let $\mathcal{G} = \{(x - \frac{\delta_x}{2}, x + \frac{\delta_x}{2}) \mid x \in X\}$. We note that $\mathcal{G}$ is an open cover of X , hence by compactness of X , there must exist a finite subcover $\mathcal{G}'$. We can write $\mathcal{G}' = \{(x_1 - \frac{\delta_{x_1}}{2}, x_1 + \frac{\delta_{x_1}}{2}), (x_2 - \frac{\delta_{x_2}}{2}, x_2 + \frac{\delta_{x_2}}{2}), \dots, (x_n - \frac{\delta_{x_n}}{2}, x_n + \frac{\delta_{x_n}}{2})\}$ for some $n \in \mathbb{N}$. Then we note that for all $x \in X$, there exists $i \in [n]$ such that $x \in (x_i - \frac{\delta_{x_i}}{2}, x_i + \frac{\delta_{x_i}}{2})$.

Now, let $\delta = \min\{\frac{\delta_{x_1}}{2}, \frac{\delta_{x_2}}{2}, \dots, \frac{\delta_{x_n}}{2}\}$. We claim that if $|x - y| < \delta$ then $|f(x) - f(y)| < \epsilon$.

We have that for all $m \in X$, if $|m - x_i| < \delta_{x_i}$ then $|f(m) - f(x_i)| < \frac{\epsilon}{2}$. Also, $|x_i - y| = |x_i - x + x - y| \leq |x_i - x| + |x - y| < \frac{\delta_{x_i}}{2} + \delta \leq \delta_{x_i}$. Hence, $|x_i - y| < \delta_{x_i}$, thus $|f(x_i) - f(y)| < \frac{\epsilon}{2}$.

Then, $|f(x) - f(y)| = |f(x) - f(x_i) + f(x_i) - f(y)| \leq |f(x) - f(x_i)| + |f(x_i) - f(y)| < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon$. This suffices to complete the proof. $\square$

Exercise 13.7. a) Prove that if $f: [0, \infty) \rightarrow \mathbb{R}$ is continuous and f is uniformly continuous on $[1, \infty)$, then f is uniformly continuous on $[0, \infty)$.

b) Prove that $f(x) = \sqrt{x}$ is uniformly continuous on $A = [0, +\infty)$.

Exercise 13.7(a.) We want to show that f is uniformly continuous on $[0, \infty)$.

Since $[0, 2]$ is compact and f is continuous, by Theorem 13.6 we have that f is uniformly continuous on $[0, 2]$. Hence, there exists $\delta_1 > 0$ such that for all $\epsilon > 0$ and for all $x, y \in [0, 2]$, if $|x - y| < \delta_1$ then $|f(x) - f(y)| < \epsilon$.

Since f is uniformly continuous on $[1, \infty)$, there exists $\delta_2 > 0$ such that for all $\epsilon > 0$ and for all $x, y \in [1, \infty)$, if $|x - y| < \delta_2$ then $|f(x) - f(y)| < \epsilon$.

Let $\delta = \min\{\delta_1, \delta_2, 1\}$, so that $\delta \leq 1$. We claim that for all $\epsilon > 0$ and for all $x, y \in [0, \infty)$, if $|x - y| < \delta$ then $|f(x) - f(y)| < \epsilon$. If both $x, y \in [1, \infty)$ or both $x, y \in [0, 2]$, then we are done by the above. Suppose without loss of generality that $x \in [0, 1)$ and $y \in (2, \infty)$. Then $|x - y| > 1$, hence $|x - y| \not< \delta$. This completes the proof. $\square$

Exercise 13.7(b.) We want to show that $f(x) = \sqrt{x}$ is uniformly continuous on $[0, \infty)$.

By Exercise 13.3(e), $\sqrt{x}$ is uniformly continuous on $[1, \infty)$. Furthermore, since $\sqrt{x}$ is continuous and $[0, 1]$ is compact, by Theorem 13.6, $\sqrt{x}$ is uniformly continuous on $[0, 1]$. Therefore, by Exercise 13.7(a), $\sqrt{x}$ is uniformly continuous on $[0, \infty)$. This completes the proof. $\square$

Corollary 13.8. *Suppose that $f: [a, b] \rightarrow \mathbb{R}$ is continuous. Then f is uniformly continuous.*

Proof of Corollary 13.8. We want to show that f is uniformly continuous on $[a, b]$. Since $[a, b]$ is compact and f is continuous, by Theorem 13.6 we have that f is uniformly continuous on $[a, b]$. This completes the proof. $\square$

Exercise 13.9. Show that if f and g are bounded on A and uniformly continuous on A , then fg is uniformly continuous on A .

Proof. Let $\epsilon > 0$. Since f is uniformly continuous on A , there exists $\delta_f > 0$ such that for all $x, y \in A$, if $|x - y| < \delta_f$ then $|f(x) - f(y)| < \frac{\epsilon}{2G}$, where G is an upper bound for $|g|$ (that is, for all $x \in A$, $|g(x)| \leq G$).

Since g is uniformly continuous on A , there exists $\delta_g > 0$ such that for all $x, y \in A$, if $|x - y| < \delta_g$ then $|g(x) - g(y)| < \frac{\epsilon}{2F}$, where F is an upper bound for $|f|$ (that is, for all $x \in A$, $|f(x)| \leq F$).

Let $\delta = \min\{\delta_f, \delta_g\}$. If $|x - y| < \delta$, then

$$\begin{aligned}
|f(x)g(x) - f(y)g(y)| &= |f(x)(g(x) - g(y)) + g(y)(f(x) - f(y))| \\
&\leq |f(x)| |g(x) - g(y)| + |g(y)| |f(x) - f(y)| \\
&\leq F \cdot |g(x) - g(y)| + G \cdot |f(x) - f(y)| \\
&< F \cdot \frac{\epsilon}{2F} + G \cdot \frac{\epsilon}{2G} \\
&= \epsilon.
\end{aligned}$$

This completes the proof. □

We are now ready to turn to integration.

Definition 13.10. Let $a < b$. A *partition* of the interval $[a, b]$ is a finite set of points in $[a, b]$ that includes a and b . We usually write partitions as $P = \{t_0, t_1, \dots, t_n\}$, with the convention that

$$a = t_0 < t_1 < \dots < t_{n-1} < t_n = b.$$

If P and Q are partitions of the interval $[a, b]$ and $P \subset Q$, we refer to Q as a *refinement* of P .

Definition 13.11. Suppose that $f: [a, b] \rightarrow \mathbb{R}$ is bounded and that $P = \{t_0, t_1, \dots, t_n\}$ is a partition of $[a, b]$. Define

$$\begin{aligned}
m_i(f) &= \inf\{f(x) \mid t_{i-1} \leq x \leq t_i\} \\
M_i(f) &= \sup\{f(x) \mid t_{i-1} \leq x \leq t_i\}.
\end{aligned}$$

The *lower sum* of f for the partition P is the number

$$L(f, P) = \sum_{i=1}^n m_i(f)(t_i - t_{i-1}).$$

The *upper sum* of f for the partition P is the number

$$U(f, P) = \sum_{i=1}^n M_i(f)(t_i - t_{i-1}).$$

Notice that it is always the case that $L(f, P) \leq U(f, P)$.

Lemma 13.12. Suppose that $f: [a, b] \rightarrow \mathbb{R}$ is bounded. Suppose that P and Q are partitions of $[a, b]$ and that Q is a refinement of P .

Then,

$$L(f, P) \leq L(f, Q) \quad \text{and} \quad U(f, P) \geq U(f, Q).$$

Proof of Lemma 13.12. Suppose that $f: [a, b] \rightarrow \mathbb{R}$ is bounded. Then, suppose that P and Q are partitions of $[a, b]$. and that Q is a refinement of P . We want to show that $L(f, P) \leq L(f, Q)$ and $U(f, P) \geq U(f, Q)$

We have that Q is a refinement of P . Suppose that $Q = P$. Then we have that $L(f, P) = L(f, Q)$ and $U(f, P) = U(f, Q)$.

Now consider the case that Q has a greater number of points than P . Then, we have that $Q = P \cup \{c\}$ where $c \in \mathbb{R}$. Then, by the definition of partitions, there exists some $[t_{i-1}, t_i]$ which contains c . We can split this interval into two separate intervals $[t_{i-1}, c]$ and $[c, t_i]$.

Then, since $m_i(f) = \inf_{[t_{i-1}, t_i]} f$, we have $f(x) \geq m_i(f)$ for all $x \in [t_{i-1}, t_i]$. In particular, since $[t_{i-1}, c] \subset [t_{i-1}, t_i]$, we have $f(x) \geq m_i(f)$ for all $x \in [t_{i-1}, c]$. Hence $m_i(f)$ is a lower bound for f on $[t_{i-1}, c]$, and since $\inf_{[t_{i-1}, c]} f$ is the greatest lower bound:

$$\inf_{[t_{i-1}, c]} f \geq m_i(f).$$

By the same argument, $f(x) \geq m_i(f)$ for all $x \in [c, t_i]$, and hence:

$$\inf_{[c, t_i]} f \geq m_i(f).$$

Therefore, since $(c - t_{i-1}) > 0$ and $(t_i - c) > 0$:

$$\inf_{[t_{i-1}, c]} f \cdot (c - t_{i-1}) + \inf_{[c, t_i]} f \cdot (t_i - c) \geq m_i(f)(c - t_{i-1}) + m_i(f)(t_i - c) = m_i(f)(t_i - t_{i-1}).$$

All other subintervals of P are unchanged in Q , so all other terms in $L(f, P)$ and $L(f, Q)$ are identical. Therefore:

$$L(f, Q) = L(f, P) - m_i(f)(t_i - t_{i-1}) + \inf_{[t_{i-1}, c]} f \cdot (c - t_{i-1}) + \inf_{[c, t_i]} f \cdot (t_i - c) \geq L(f, P).$$

The argument for $U(f, Q) \leq U(f, P)$ is analogous: since $[t_{i-1}, c] \subset [t_{i-1}, t_i]$ and $[c, t_i] \subset [t_{i-1}, t_i]$, we have $\sup_{[t_{i-1}, c]} f \leq M_i(f)$ and $\sup_{[c, t_i]} f \leq M_i(f)$ by the same subset argument, and hence $U(f, Q) \leq U(f, P)$. □

Theorem 13.13. *Let P_1 and P_2 be partitions of $[a, b]$ and suppose that $f: [a, b] \rightarrow \mathbb{R}$ is bounded. Then,*

$$L(f, P_1) \leq U(f, P_2).$$

Proof of Theorem 13.13. Let P_1 and P_2 be partitions of $[a, b]$. Suppose that $f: [a, b] \rightarrow \mathbb{R}$ is bounded. We want to show that $L(f, P_1) \leq U(f, P_2)$.

Let $R = P_1 \cup P_2$. Then, we have that R is a refinement of both P_1 and P_2 . Therefore, by **Lemma 13.12** we have that

$$L(f, P_1) \leq L(f, R) \quad \text{and} \quad U(f, P_2) \geq U(f, R)$$

Therefore, we have that $L(f, P_1) \leq L(f, R) \leq U(f, R) \leq U(f, P_2)$. Thus, we have that $L(f, P_1) \leq U(f, P_2)$. This completes the proof. $\square$

Definition 13.14. Let $f: [a, b] \rightarrow \mathbb{R}$ be bounded. We define:

$$L(f) = \sup\{L(f, P) \mid P \text{ is a partition of } [a, b]\}$$

$$U(f) = \inf\{U(f, P) \mid P \text{ is a partition of } [a, b]\}$$

to be, respectively, the *lower integral* and *upper integral* of f from a to b .

Exercise 13.15. Why do $L(f)$ and $U(f)$ exist? Find a function f for which $L(f) = U(f)$. Find a function f for which $L(f) \neq U(f)$. Prove that $L(f) \leq U(f)$.

Exercise 13.15. Define set $J = \{L(f, P) \mid P \text{ is a partition of } [a, b]\}$. The set $\{a, b\}$ is a partition of $[a, b]$, hence J is non-empty.

We want to show that J is bounded above. Fix M to be any partition of $[a, b]$, then by Theorem 13.13, we have that $U(f, M)$ is an upper bound for J . Hence, J is non-empty and bounded above, hence $\sup J$ exists, hence $L(f)$ exists. Analogous reasoning holds for $U(f)$.

Consider the function $f: [a, b] \rightarrow \mathbb{R}$ defined as $f(x) = k$. Then, $m_i(f) = M_i(f) = k$, hence for every partition P , we have that $L(f, P) = U(f, P) = k(t_n - t_0) = k(b - a)$, hence $L(f) = U(f) = k(b - a)$.

Consider the function $f: [a, b] \rightarrow \mathbb{R}$ defined as

$$f(x) = \begin{cases} 0 & \text{if } x \in \mathbb{Q} \\ 1 & \text{if } x \in \mathbb{R} \setminus \mathbb{Q}. \end{cases}$$

For all $t_1, t_2 \in [a, b]$, there exist $x, y \in [t_1, t_2]$ such that $f(x) = 0$ and $f(y) = 1$ (since both $\mathbb{Q}$ and $\mathbb{R} \setminus \mathbb{Q}$ are dense in $\mathbb{R}$).

Hence, for all partitions P of $[a, b]$, we have that $m_i = 0$ and $M_i = 1$. Hence, for all partitions P of $[a, b]$, we have that $L(f, P) = 0$ and $U(f, P) = \sum_{i=1}^n 1 \cdot (t_i - t_{i-1}) = t_n - t_0 = b - a$.

Hence, $L(f) = 0$ and $U(f) = b - a$, thus $L(f) \neq U(f)$.

We have that $L(f, P_1) \leq U(f, P_2)$ for all P_1, P_2 . Hence, fix a certain P_2 , then we note that $U(f, P_2)$ is an upper bound for $\{L(f, P_1) \mid P_1 \text{ is a partition of } [a, b]\}$. Then, by the definition of supremum, $L(f) = \sup\{L(f, P_1) \mid P_1 \text{ is a partition of } [a, b]\} \leq U(f, P_2)$. This means that $L(f)$ is a lower bound of $U(f, P_2)$. Then, by the definition of infimum, $U(f) = \inf\{U(f, P_2) \mid P_2 \text{ is a partition of } [a, b]\} \geq L(f)$. Therefore, $L(f) \leq U(f)$.

This completes the proof. $\square$

Definition 13.16. Let $f: [a, b] \rightarrow \mathbb{R}$ be bounded. We say that f is *integrable* on $[a, b]$ if $L(f) = U(f)$. In this case, the common value $L(f) = U(f)$ is called the *integral* of f from a to b and we write it as:

$$\int_a^b f.$$

Note that if f is an integrable function on $[a, b]$, it is necessarily bounded.

When we want to display the variable of integration, we write the integral as follows, including the symbol dx to indicate that variable of integration:

$$\int_a^b f(x) dx.$$

For example, if $f(x) = x^2$, we would write $\int_a^b x^2 dx$ but not $\int_a^b x^2$.

Exercise 13.17. Fix $c \in \mathbb{R}$ and let $f: [a, b] \rightarrow \mathbb{R}$ be defined by $f(x) = c$, for each $x \in [a, b]$. Show that f is integrable on $[a, b]$ and that $\int_a^b f = c(b - a)$.

Exercise 13.17. Fix $c \in \mathbb{R}$ and let $f: [a, b] \rightarrow \mathbb{R}$ be defined by $f(x) = c$, for each $x \in [a, b]$. We want to show that f is integrable on $[a, b]$ and that $\int_a^b f = c(b - a)$.

To show that f is integrable on $[a, b]$, it suffices to show that $L(f) = U(f)$. Consider any arbitrary partition P of $[a, b]$. As $f(x) = c$, we have that $\forall i \in [n]$, $m_i(f) = \inf\{f(x) \mid t_{i-1} \leq x \leq t_i\} = c$. Similarly, we have that $\forall i \in [n]$, $M_i(f) = \sup\{f(x) \mid t_{i-1} \leq x \leq t_i\} = c$.

Now consider $L(f, P)$ and $U(f, P)$. By **Definition 13.11**, we have that

$$L(f, P) = \sum_{i=1}^n m_i(f)(t_{i-1} - t_i) = m_i(f)(b - a) = c \cdot (b - a)$$

Similarly, we have that

$$U(f, P) = \sum_{i=1}^n M_i(f)(t_{i-1} - t_i) = M_i(f)(b - a) = c \cdot (b - a)$$

Therefore, we have that $L(f, P) = c \cdot (b - a) = U(f, P)$ for an arbitrary partition P . As the partition is arbitrary, we have that this applies for all partitions P of $[a, b]$. Therefore, we have that $L(f) = c \cdot (b - a) = U(f)$. Thus, this suffices to show that f is integrable and $\int_a^b f = c \cdot (b - a)$. $\square$

Theorem 13.18. Let $f: [a, b] \rightarrow \mathbb{R}$ be bounded. Then f is integrable if and only if for every $\epsilon > 0$ there exists a partition P of $[a, b]$ such that

$$U(f, P) - L(f, P) < \epsilon.$$

Proof of Theorem 13.18. ($\Rightarrow$) Suppose that f is integrable, so $L(f) = U(f)$. Let $\epsilon > 0$. By the definition of supremum and infimum, there exist partitions P_1 and P_2 of $[a, b]$ such that:

$$L(f, P_1) > L(f) - \frac{\epsilon}{2} \quad \text{and} \quad U(f, P_2) < U(f) + \frac{\epsilon}{2}.$$

Let $P = P_1 \cup P_2$ be their finest common refinement. By Lemma 13.12:

$$L(f, P_1) \leq L(f, P) \quad \text{and} \quad U(f, P) \leq U(f, P_2).$$

Therefore:

$$U(f, P) - L(f, P) \leq U(f, P_2) - L(f, P_1) < U(f) + \frac{\epsilon}{2} - L(f) + \frac{\epsilon}{2} = \epsilon,$$

where the last equality uses $L(f) = U(f)$.

($\Leftarrow$) Suppose that for every $\epsilon > 0$ there exists a partition P of $[a, b]$ such that $U(f, P) - L(f, P) < \epsilon$. Since $L(f, P) \leq L(f) \leq U(f) \leq U(f, P)$, we have:

$$0 \leq U(f) - L(f) \leq U(f, P) - L(f, P) < \epsilon.$$

Since this holds for all $\epsilon > 0$ and $U(f) - L(f) \geq 0$, we conclude $U(f) - L(f) = 0$, and hence $L(f) = U(f)$. Therefore f is integrable. $\square$

Theorem 13.19. *If $f: [a, b] \rightarrow \mathbb{R}$ is continuous, then f is integrable.*
(Hint: Use Theorem 13.18 and uniform continuity.)

Proof of Theorem 13.19. Let $f: [a, b] \rightarrow \mathbb{R}$ be continuous. We want to show that f is integrable.

Let $f: [a, b] \rightarrow \mathbb{R}$ be continuous. Then, as $[a, b]$ is compact by **Theorem 10.16**, we have that f is uniformly continuous by **Theorem 13.6**. Therefore, we have that $\forall \epsilon > 0, \exists \delta > 0$ such that $\forall x, y \in [a, b]$ if $|x - y| < \delta$ then $|f(x) - f(y)| < \epsilon$.

Now, consider an arbitrary partition P of $[a, b]$ such that $P = \{t_0, t_1, t_2, \dots, t_n\}$ for some $n \in \mathbb{N}$. Then as f is uniformly continuous we have that $\forall \epsilon > 0, \exists \delta > 0$ such that $\forall t_{i-1}, t_i \in [a, b]$ if $|t_i - t_{i-1}| < \delta$ then $|f(t_i) - f(t_{i-1})| < \epsilon$. Let $\delta = t_i - t_{i-1}$. Then, $\forall t_i, t_{i-1} \in [a, b]$ we have that $|t_i - t_{i-1}| < \delta$ for all $i \in [n]$ and, as f is uniformly continuous across $[a, b]$, we have that $|f(t_i) - f(t_{i-1})| < \epsilon$.

Then, fix $\epsilon' = \frac{\epsilon}{b-a}$. Then, consider $M_i(f) - m_i(f)$. We have that this is the greatest spread between $f(t_{i-1})$ and $f(t_i)$. Hence,

$$\sup_{\forall t_i, t_{i-1} \in [a, b]} |f(t_i) - f(t_{i-1})| < \epsilon' = \frac{\epsilon}{b-a}$$

This is equivalent to

$$|M_i(f) - m_i(f)| < \epsilon' = \frac{\epsilon}{b-a}$$

Now, we want to show that $U(f, P) - L(f, P) < \epsilon$. By **Definition 13.11**, we have that

$$U(f, P) - L(f, P) = \sum_{i=1}^n [M_i(f)(t_i - t_{i-1}) - m_i(f)(t_i - t_{i-1})] = \sum_{i=1}^n (t_i - t_{i-1})[M_i(f) - m_i(f)]$$

As $|M_i(f) - m_i(f)| < \epsilon' = \frac{\epsilon}{b-a}$, we have that

$$\sum_{i=1}^n (t_i - t_{i-1})[M_i(f) - m_i(f)] < (b-a) \cdot \frac{\epsilon}{b-a}$$

Thus, this suffices to complete the proof. □

Lemma 13.20. *Let $f: [a, b] \rightarrow \mathbb{R}$ be bounded. Given $\Omega \in \mathbb{R}$, we have that $\Omega = \int_a^b f$ if, and only if, for all $\epsilon > 0$, there is some partition P such that*

$$U(f, P) - \Omega < \epsilon \quad \text{and} \quad \Omega - L(f, P) < \epsilon.$$

Proof of Lemma 13.20. ($\Rightarrow$) Suppose that $\Omega = \int_a^b f$. We want to show that for all $\epsilon > 0$ there exists a partition P_ϵ such that $U(f, P_\epsilon) - \Omega < \epsilon$ and $\Omega - L(f, P_\epsilon) < \epsilon$.

Since $\Omega = \int_a^b f$, we have $L(f) = \Omega = U(f)$, and hence we have that

$$\sup_P L(f, P) = \Omega = \inf_P U(f, P).$$

By the definition of supremum and infimum, there exist partitions P_1 and P_2 such that

$$L(f, P_1) > \Omega - \epsilon \quad \text{and} \quad U(f, P_2) < \Omega + \epsilon.$$

Now, let $P_\epsilon = P_1 \cup P_2$ be their finest common refinement. Since P_ϵ is a refinement of both P_1 and P_2 , by **Lemma 13.12**:

$$\Omega - \epsilon < L(f, P_1) \leq L(f, P_\epsilon) \quad \text{and} \quad U(f, P_\epsilon) \leq U(f, P_2) < \Omega + \epsilon.$$

Then, rearranging this gives $\Omega - L(f, P_\epsilon) < \epsilon$ and $U(f, P_\epsilon) - \Omega < \epsilon$, as required.

($\Leftarrow$) Suppose that for all $\epsilon > 0$ there exists a partition P_ϵ such that $U(f, P_\epsilon) - \Omega < \epsilon$ and $\Omega - L(f, P_\epsilon) < \epsilon$. We want to show that $\Omega = \int_a^b f$. In other words, we want to show that $L(f) = \Omega = U(f)$.

Since $U(f) = \inf_P U(f, P) \leq U(f, P_\epsilon)$ and $L(f, P_\epsilon) \leq \sup_P L(f, P) = L(f)$, we have:

$$\Omega - \epsilon < L(f, P_\epsilon) \leq L(f) \leq U(f) \leq U(f, P_\epsilon) < \Omega + \epsilon.$$

Hence $\Omega - \epsilon < L(f) \leq U(f) < \Omega + \epsilon$ for all $\epsilon > 0$.

We now show $L(f) = \Omega = U(f)$ by contradiction.

Case 1: Suppose $L(f) < \Omega$. Then there exists $\delta > 0$ such that $L(f) \leq \Omega - \delta$. Pick $\epsilon = \frac{\delta}{2}$. Then:

$$\Omega - \epsilon = \Omega - \frac{\delta}{2} > \Omega - \delta \geq L(f),$$

which contradicts $\Omega - \epsilon < L(f)$.

Case 2: Suppose $\Omega < U(f)$. Then there exists $\delta > 0$ such that $U(f) \geq \Omega + \delta$. Pick $\epsilon = \frac{\delta}{2}$. Then:

$$\Omega + \epsilon = \Omega + \frac{\delta}{2} < \Omega + \delta \leq U(f),$$

which contradicts $U(f) < \Omega + \epsilon$.

In both cases we reach a contradiction, so $L(f) = \Omega = U(f)$, and therefore $\Omega = \int_a^b f$. This completes the proof. $\square$

Exercise 13.21. Define $f: [0, b] \rightarrow \mathbb{R}$ by the formula $f(x) = x$. Show that f is integrable on $[0, b]$ and that $\int_0^b f = \frac{b^2}{2}$. *Hint:* $\sum_{i=1}^n i = \frac{1}{2}n(n+1)$.

Exercise 13.21. Define $f: [0, b] \rightarrow \mathbb{R}$ as $f(x) = x$. We want to first show that f is integrable on $[0, b]$.

We have that $f(x) = x$ is continuous. Therefore, we have that $[0, b] \subset \mathbb{R}$ is compact and that $f(x)$ is continuous. Therefore, we have that f is integrable on $[0, b]$.

Secondly, we want to show that $\int_0^b f = \frac{b^2}{2}$. Consider the uniform partition $P_n = \{0, \frac{b}{n}, \frac{2b}{n}, \dots, b\}$ with n equal subintervals of width $\frac{b}{n}$. As $f(x) = x$ is increasing, we have that $M_i(f) = \sup\{f(x) \mid t_{i-1} \leq x \leq t_i\} = t_i = \frac{ib}{n}$. Therefore:

$$U(f, P_n) = \sum_{i=1}^n M_i(f)(t_i - t_{i-1}) = \sum_{i=1}^n \frac{ib}{n} \cdot \frac{b}{n} = \frac{b^2}{n^2} \sum_{i=1}^n i = \frac{b^2}{n^2} \cdot \frac{n(n+1)}{2} = \frac{b^2(n+1)}{2n}.$$

Similarly, as $m_i(f) = t_{i-1} = \frac{(i-1)b}{n}$, we have that:

$$L(f, P_n) = \sum_{i=1}^n m_i(f)(t_i - t_{i-1}) = \frac{b^2}{n^2} \sum_{i=1}^n (i-1) = \frac{b^2}{n^2} \cdot \frac{n(n-1)}{2} = \frac{b^2(n-1)}{2n}.$$

By **Lemma 13.20**, it suffices to show that for all $\epsilon > 0$ there exists a partition P such that $U(f, P) - \frac{b^2}{2} < \epsilon$ and $\frac{b^2}{2} - L(f, P) < \epsilon$. We have that:

$$U(f, P_n) - \frac{b^2}{2} = \frac{b^2(n+1)}{2n} - \frac{b^2}{2} = \frac{b^2}{2n},$$

$$\frac{b^2}{2} - L(f, P_n) = \frac{b^2}{2} - \frac{b^2(n-1)}{2n} = \frac{b^2}{2n}.$$

For any $\epsilon > 0$, pick $n > \frac{b^2}{2\epsilon}$. Then $\frac{b^2}{2n} < \epsilon$, and hence by **Lemma 13.20**, $\int_0^b f = \frac{b^2}{2}$. $\square$

Exercise 13.22. Show that the converse of Theorem 13.19 is false in general.

Exercise 13.22. Let f be integrable. We want to show that f is not continuous.

Consider the function

$$f(x) = \begin{cases} 1 & \text{if } x \geq 0 \\ 0 & \text{if } x < 0 \end{cases}$$

Firstly, we want to show that f is integrable. For any $\epsilon > 0$, define $P_\epsilon = \{-1, -\frac{\epsilon}{2}, 0, 1\}$. Then

$$L(f, P_\epsilon) = 0 \cdot \left(-\frac{\epsilon}{2} + 1\right) + 0 \cdot \frac{\epsilon}{2} + 1 \cdot 1 = 1$$

$$U(f, P_\epsilon) = 0 \cdot \left(-\frac{\epsilon}{2} + 1\right) + 1 \cdot \frac{\epsilon}{2} + 1 \cdot 1 = 1 + \frac{\epsilon}{2}.$$

Hence $U(f, P_\epsilon) - L(f, P_\epsilon) = \frac{\epsilon}{2} < \epsilon$. By **Theorem 13.18**, f is integrable on $[-1, 1]$. Hence, we have that $f(x)$ is integrable.

Secondly, we want to show that $f(x)$ is not continuous. In other words, we want to show that there exists $\epsilon > 0$ such that there does not exist $\delta > 0$ where if $x, y \in \mathbb{R}$ and $|x - y| < \delta$ then $|f(x) - f(y)| < \epsilon$. Choose $\epsilon = \frac{1}{2}$. For any $\delta > 0$, take $x = \frac{\delta}{2} > 0$ and $y = -\frac{\delta}{2} < 0$. Then $|x - y| = \delta$ but $|f(x) - f(y)| = |1 - 0| = 1 > \frac{1}{2} = \epsilon$.

Thus this suffices to prove that f is not continuous at $x = 0$, and the converse of Theorem 13.19 is false $\square$

Theorem 13.23. Let $a < b < c$. A function $f: [a, c] \rightarrow \mathbb{R}$ is integrable on $[a, c]$ if and only if f is integrable on $[a, b]$ and $[b, c]$. When f is integrable on $[a, c]$, we have

$$\int_a^c f = \int_a^b f + \int_b^c f.$$

Proof of Theorem 13.23. Let $a < b < c$. We want to show that $f: [a, c] \rightarrow \mathbb{R}$ is integrable on $[a, c]$ if and only if f is integrable on $[a, b]$ and $[b, c]$.

($\Rightarrow$) Let $f: [a, c] \rightarrow \mathbb{R}$ be integrable on $[a, c]$. By **Theorem 13.18**, for every $\epsilon > 0$ there exists a partition P of $[a, c]$ such that $U(f, P) - L(f, P) < \epsilon$. Let $P' = P \cup \{b\}$. By the nature of the interval we can split P' into $P_{AB} = P' \cap [a, b]$ and $P_{BC} = P' \cap [b, c]$. Then

$$\begin{aligned} \sum_{i=1}^{n_{AB}} m_i(f)(t_i - t_{i-1}) + \sum_{i=1}^{n_{BC}} m_i(f)(t_i - t_{i-1}) &= \sum_{i=1}^{n'} m_i(f)(t_i - t_{i-1}) \\ \sum_{i=1}^{n_{AB}} M_i(f)(t_i - t_{i-1}) + \sum_{i=1}^{n_{BC}} M_i(f)(t_i - t_{i-1}) &= \sum_{i=1}^{n'} M_i(f)(t_i - t_{i-1}) \end{aligned}$$

and so $U(f, P_{AB}) - L(f, P_{AB}) \leq U(f, P') - L(f, P') \leq U(f, P) - L(f, P) < \epsilon$ by **Lemma 13.12**. This argument is symmetric for P_{BC} . By **Theorem 13.18**, f is integrable on $[a, b]$ and $[b, c]$.

($\Leftarrow$) Suppose f is integrable on $[a, b]$ and $[b, c]$. Let $\epsilon > 0$. Then there exist partitions P_1 of $[a, b]$ and P_2 of $[b, c]$ such that

$$U(f, P_1) - L(f, P_1) < \frac{\epsilon}{2} \quad \text{and} \quad U(f, P_2) - L(f, P_2) < \frac{\epsilon}{2}.$$

Let $P = P_1 \cup P_2$, which is a partition of $[a, c]$. Then

$$U(f, P) - L(f, P) = (U(f, P_1) - L(f, P_1)) + (U(f, P_2) - L(f, P_2)) < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon.$$

By **Theorem 13.18**, f is integrable on $[a, c]$. □

Remark 13.24. If $a = b$, we define

$$\int_a^b f = \int_a^a f = 0;$$

if $b < a$, we define

$$\int_a^b f = - \int_b^a f,$$

whenever the latter integral exists. With this notational convention, it follows that the equation

$$\int_a^c f = \int_a^b f + \int_b^c f$$

always holds, regardless of the ordering of a , b and c , whenever f is integrable on the largest of the three intervals.

Theorem 13.25. *Suppose that f and g are integrable functions on $[a, b]$ and that $c \in \mathbb{R}$ is a constant. Then $f + g$ and cf are integrable on $[a, b]$ and*

$$(a) \quad \int_a^b (f + g) = \int_a^b f + \int_a^b g, \text{ and}$$

$$(b) \quad \int_a^b c \cdot f = c \int_a^b f.$$

Proof of Theorem 13.25(a.) Let f and g be integrable on $[a, b]$. We want to show that $f + g$ is integrable and $\int_a^b (f + g) = \int_a^b f + \int_a^b g$.

Let $\Omega = \int_a^b f + \int_a^b g$. By **Lemma 13.20**, it suffices to show that for all $\epsilon > 0$, there exists a partition P such that

$$U(f + g, P) - \Omega < \epsilon \quad \text{and} \quad \Omega - L(f + g, P) < \epsilon.$$

Let $\epsilon > 0$. Since f and g are integrable, there exist partitions P_1 and P_2 of $[a, b]$ such that

$$U(f, P_1) - L(f, P_1) < \frac{\epsilon}{2} \quad \text{and} \quad U(g, P_2) - L(g, P_2) < \frac{\epsilon}{2}.$$

Let $P = P_1 \cup P_2$. By **Lemma 13.12**, refining a partition does not increase upper sums or decrease lower sums. Since $m_i(f) + m_i(g) \leq m_i(f + g) \leq M_i(f + g) \leq M_i(f) + M_i(g)$, we have

$$L(f, P) + L(g, P) \leq L(f + g, P) \leq U(f + g, P) \leq U(f, P) + U(g, P).$$

Now, since $L(f, P) \leq \int_a^b f \leq U(f, P)$ and $L(g, P) \leq \int_a^b g \leq U(g, P)$, we have

$$U(f + g, P) - \Omega \leq U(f, P) + U(g, P) - \int_a^b f - \int_a^b g < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon$$

$$\Omega - L(f + g, P) \leq \int_a^b f + \int_a^b g - L(f, P) - L(g, P) < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon.$$

By **Lemma 13.20**, $f + g$ is integrable and $\int_a^b (f + g) = \int_a^b f + \int_a^b g$. □

Proof of Theorem 13.25(b.) Let f be integrable on $[a, b]$ and $c \in \mathbb{R}$. We want to show that cf is integrable and $\int_a^b c \cdot f = c \int_a^b f$.

Let $\Omega = c \int_a^b f$. By **Lemma 13.20**, it suffices to show that for all $\epsilon > 0$, there exists a partition P such that

$$U(cf, P) - \Omega < \epsilon \quad \text{and} \quad \Omega - L(cf, P) < \epsilon.$$

Let $\epsilon > 0$. Since f is integrable, there exists a partition P of $[a, b]$ such that

$$U(f, P) - L(f, P) < \frac{\epsilon}{|c|}$$

where if $c = 0$ the result is immediate since $cf = 0$ is integrable with $\int_a^b cf = 0 = c \int_a^b f$.

For $c > 0$, we have $m_i(cf) = c \cdot m_i(f)$ and $M_i(cf) = c \cdot M_i(f)$, so

$$L(cf, P) = c \cdot L(f, P) \quad \text{and} \quad U(cf, P) = c \cdot U(f, P).$$

For $c < 0$, the inequality flips, so $m_i(cf) = c \cdot M_i(f)$ and $M_i(cf) = c \cdot m_i(f)$, giving

$$L(cf, P) = c \cdot U(f, P) \quad \text{and} \quad U(cf, P) = c \cdot L(f, P).$$

In both cases, $U(cf, P) - L(cf, P) = |c| \cdot (U(f, P) - L(f, P)) < \epsilon$. Moreover,

$$U(cf, P) - \Omega \leq U(cf, P) - L(cf, P) < \epsilon \quad \text{and} \quad \Omega - L(cf, P) \leq U(cf, P) - L(cf, P) < \epsilon.$$

By **Lemma 13.20**, cf is integrable and $\int_a^b c \cdot f = c \int_a^b f$. □

Theorem 13.26. Suppose that f and g are integrable functions on $[a, b]$ with $f(x) \leq g(x)$, for all $x \in [a, b]$. Then

$$\int_a^b f \leq \int_a^b g.$$

Proof of Theorem 13.26. Suppose that f and g are integrable functions on $[a, b]$ with $f(x) \leq g(x)$, for all $x \in [a, b]$. We want to show that $\int_a^b f \leq \int_a^b g$.

Let $h(x) = g(x) - f(x)$ for all $x \in [a, b]$. As f and g are integrable, by **Theorem 13.25(a.)**, we have that h is integrable and, as $f(x) \leq g(x) \forall x \in [a, b]$, $h(x) \geq 0 \forall x \in [a, b]$. For any partition P , we have that $m_i(h) \geq 0$ for all i , hence $L(h, P) \geq 0$ and thus $\int_a^b h \geq 0$.

As $\int_a^b h \geq 0$, we have that $\int_a^b (g - f) \geq 0$ and equivalently $\int_a^b g - \int_a^b f \geq 0$. Thus, this suffices to prove that $\int_a^b g \geq \int_a^b f$. □

Theorem 13.27. Suppose that f is an integrable function on $[a, b]$. Then $|f|$ is also integrable on $[a, b]$ and

$$\left| \int_a^b f \right| \leq \int_a^b |f|.$$

Proof of Theorem 13.27. Define $f^+, f^- : [a, b] \rightarrow \mathbb{R}$ as $f^+(x) = \max\{f(x), 0\}$ and $f^-(x) = \min\{f(x), 0\}$. We note that $f = f^+ + f^-$ and $|f| = f^+ - f^-$. We want to show that $|f|$ is integrable and that $\left| \int_a^b f \right| \leq \int_a^b |f|$.

We claim that $M_i(f^+) - m_i(f^+) \leq M_i(f) - m_i(f)$. We consider two cases.

Case 1: If $M_i(f) \leq 0$ then $f^+ = 0$, hence $M_i(f^+) = m_i(f^+) = 0$. Thus, we have that $M_i(f^+) - m_i(f^+) = 0 \leq M_i(f) - m_i(f)$.

Case 2: If $M_i(f) > 0$, then $M_i(f^+) = M_i(f)$ and $m_i(f^+) \geq 0$, which implies $m_i(f^+) \geq m_i(f)$. Thus, we have that $M_i(f^+) - m_i(f^+) \leq M_i(f) - m_i(f)$.

Analogously, we have that $M_i(f^-) - m_i(f^-) \leq M_i(f) - m_i(f)$.

As f is integrable, we have that for all $\epsilon > 0$ there exists a partition P of $[a, b]$ such that $U(f, P) - L(f, P) < \epsilon$. Then, as $U(f^+, P) - L(f^+, P) \leq U(f, P) - L(f, P) < \epsilon$, we have that f^+ is integrable. Similarly, as $U(f^-, P) - L(f^-, P) \leq U(f, P) - L(f, P) < \epsilon$, we have that f^- is integrable. Thus, we have that $|f| = f^+ - f^-$ is integrable.

Further, we note that

$$\left| \int_a^b f \right| = \left| \int_a^b f^+ + \int_a^b f^- \right| \leq \left| \int_a^b f^+ \right| + \left| \int_a^b f^- \right| = \int_a^b f^+ - \int_a^b f^- = \int_a^b |f|.$$

This suffices to prove that $\left| \int_a^b f \right| \leq \int_a^b |f|$. This completes the proof. □

Theorem 13.28. Suppose that f is integrable on $[a, b]$ and $m \leq f(x) \leq M$, for all $x \in [a, b]$. Then

$$m(b-a) \leq \int_a^b f \leq M(b-a).$$

Proof of Theorem 13.28. Suppose that f is integrable on $[a, b]$ and $m \leq f(x) \leq M$, for all $x \in [a, b]$. We want to show $m(b-a) \leq \int_a^b f \leq M(b-a)$.

Let $g(x)$ be a function defined as $g(x) = m$. By **Exercise 13.17**, we have that $g(x)$ is integrable on $[a, b]$ where $\int_a^b g(x) = m \cdot (b-a)$. Then, as $m \leq f(x)$ for all $x \in [a, b]$ we have that $g(x) \leq f(x)$ and, by consequence $\int_a^b g(x) \leq \int_a^b f(x)$. Therefore, as $\int_a^b g(x) = m \cdot (b-a)$ we have that $m(b-a) \leq \int_a^b f$.

Now, let $h(x)$ be a function defined as $h(x) = M$. Then, similarly, we have that $h(x)$ is integrable on $[a, b]$ where $\int_a^b h(x) = M \cdot (b-a)$. Then, we have $f(x) \leq M$ for all $x \in [a, b]$. Therefore, we have that as $f(x) \leq M$ for all $x \in [a, b]$ we have that $f(x) \leq h(x)$ and, by consequence $\int_a^b f(x) \leq \int_a^b h(x)$. Therefore, as $\int_a^b h(x) = M \cdot (b-a)$ we have that $\int_a^b f \leq M \cdot (b-a)$.

Thus, through the two statements, we have that $m(b-a) \leq \int_a^b f \leq M(b-a)$. This suffices to complete the proof. $\square$

Theorem 13.29. Suppose that f is integrable on $[a, b]$. Define $F: [a, b] \rightarrow \mathbb{R}$ by

$$F(x) = \int_a^x f.$$

Then F is continuous.

Proof of Theorem 13.29. Suppose that f is integrable on $[a, b]$ and define $F: [a, b] \rightarrow \mathbb{R}$ by $F(x) = \int_a^x f$. We want to show that F is continuous.

To show that F is continuous it suffices to show that for all $\epsilon > 0$ there exists $\delta > 0$ such that if $x, y \in [a, b]$ and $|x - y| < \delta$ then $|F(x) - F(y)| < \epsilon$.

Let $\epsilon > 0$ and let u be an upper bound of f . Then, let $\delta = \frac{\epsilon}{u+1}$. Now, let $x, y \in [a, b]$ such that $|x - y| < \delta$. As $x, y \in [a, b]$ we have that this is always possible.

Now, consider $|F(x) - F(y)|$. Then, as $F(x) = \int_a^x f$ and F is integrable, we have that $|F(x) - F(y)| = |\int_a^x f - \int_a^y f|$. Now, consider the two cases

1. Let $x > y$. Then, we have that $|\int_a^x f - \int_a^y f| = |(\int_y^x f + \int_a^y f) - \int_a^y f| = |\int_y^x f|$.
2. Let $y > x$. Then, we have that $|\int_a^x f - \int_a^y f| = |\int_a^x f - (\int_x^y f + \int_a^x f)| = |-\int_x^y f| = |\int_y^x f|$.

Now, consider $|\int_y^x f|$. We have that $|\int_y^x f| \leq \int_y^x |f|$. Then, as u is an upper bound, we can write this as $\int_y^x |f| \leq u|x - y|$. As $|x - y| < \delta$, we have that $\int_y^x |f| \leq u|x - y| < u\delta = u \cdot \frac{\epsilon}{u+1} < \epsilon$. Thus, as this works for any, $\epsilon > 0$, this suffices to show that F is continuous. $\square$

Additional Exercises

1. a) Prove that if A and B are bounded nonempty sets with $A \subset B$ then

$$\inf(B) \leq \inf A \leq \sup A \leq \sup B.$$

- b) Assume that A and B are non-empty bounded sets. Let $A + B = \{a + b \mid a \in A, b \in B\}$. Prove that

$$\sup(A + B) = \sup A + \sup B.$$

- c) Let $A \subset \mathbb{R}$ be a non-empty set and let $f : A \rightarrow \mathbb{R}$ be a bounded function. Show that

$$\sup\{f(x) \mid x \in A\} - \inf\{f(x) \mid x \in A\} = \sup\{f(x) - f(y) \mid x, y \in A\}.$$

Additional Exercise 1(a.) Let A and B be bounded nonempty sets such that $A \subset B$. We want to show that $\inf(B) \leq \inf A \leq \sup A \leq \sup B$.

Let $L_A = \{x \in \mathbb{R} \mid x \leq a \text{ for all } a \in A\}$ be the set of lower bounds of A , and let $L_B = \{x \in \mathbb{R} \mid x \leq b \text{ for all } b \in B\}$ be the set of lower bounds of B . Since $A \subset B$, any lower bound of B is also a lower bound of A , and hence $L_B \subset L_A$. Now, $\inf B = \max L_B$ and $\inf A = \max L_A$. Since $L_B \subset L_A$, we have $\max L_B \leq \max L_A$, and therefore $\inf B \leq \inf A$.

The inequality $\inf A \leq \sup A$ follows from the fact that A is nonempty: pick any $a \in A$, then $\inf A \leq a \leq \sup A$.

By a symmetric argument, let $U_A = \{x \in \mathbb{R} \mid x \geq a \text{ for all } a \in A\}$ and $U_B = \{x \in \mathbb{R} \mid x \geq b \text{ for all } b \in B\}$ be the sets of upper bounds of A and B respectively. Since $A \subset B$, any upper bound of B is also an upper bound of A , and hence $U_B \subset U_A$. Therefore $\sup B = \min U_B \geq \min U_A = \sup A$, giving $\sup A \leq \sup B$.

□

Additional Exercise 1(b.) Let A and B be non-empty bounded sets. Let $A + B$ be defined as $\{a + b \mid a \in A, b \in B\}$. We want to show that $\sup(A + B) = \sup A + \sup B$.

Consider $\sup A$ and $\sup B$. We want to show that $\sup A + \sup B$ is an upper bound of $A + B$. Let $n = \sup A + \sup B$. By definition, we have that $\forall a \in A, \sup A \geq a$ and $\forall b \in B, \sup B \geq b$. As $\sup A + \sup B$ is the sum of the suprema of both sets, it follows that $\forall x \in (A + B)$ we have that $\sup A + \sup B \geq x$. Therefore, $\sup A + \sup B$ is an upper bound.

Now, we want to show that $\sup A + \sup B = \sup(A + B)$. Suppose for contradiction that M is an upper bound of $A + B$ with $M < \sup A + \sup B$. Let $\varepsilon = \sup A + \sup B - M > 0$. By the definition of supremum, there exists $a \in A$ such that $a > \sup A - \frac{\varepsilon}{2}$, and there exists $b \in B$ such that $b > \sup B - \frac{\varepsilon}{2}$. Adding these:

$$a + b > \sup A + \sup B - \varepsilon = M.$$

Hence $a + b \in A + B$ and $a + b > M$, contradicting M being an upper bound of $A + B$. Therefore $\sup(A + B) = \sup A + \sup B$. $\square$

Additional Exercise 1(c). Let $F = \{f(x) | x \in A\}$ and let $D = \{f(x) - f(y) | x, y \in A\}$. We want to show that $\sup F - \inf F = \sup D$.

First, we want to show that $\sup F - \inf F$ is an upper bound for D . Let $d \in D$, then we have that $d = f(x) - f(y)$ for some $x, y \in A$. We note that $f(x) \leq \sup F, f(y) \geq \inf F$ hence $d = f(x) - f(y) \leq \sup F - \inf F$.

Next, let m be an upper bound for D , then we want to show that $m \geq \sup F - \inf F$. Suppose for sake of contradiction that $m < \sup F - \inf F$.

Then, let $k = \sup F - \inf F - m$ and we note that $k > 0$. Since $\sup F - \frac{k}{2} < \sup F$, we note that $\sup F - \frac{k}{2}$ cannot be an upper bound for F , hence there exists $f_0 \in F$ such that $f_0 > \sup F - \frac{k}{2}$.

Similarly, since $\inf F + \frac{k}{2} > \inf F$, we note that $\inf F + \frac{k}{2}$ cannot be a lower bound for F , hence there exists $f_1 \in F$ such that $f_1 < \inf F + \frac{k}{2}$.

Then, $f_0 - f_1 > \sup F - \frac{k}{2} - \inf F - \frac{k}{2} = \sup F - \inf F - k = m$ and we note that $f_0 - f_1 \in D$. This contradicts the assumption that m is an upper bound for D .

This completes the proof. $\square$

2. Suppose that $|f(x) - g(x)| \leq \frac{\varepsilon}{2}$ for all $x \in [a, b]$. Let $M_f = \sup\{f(x) | x \in [a, b]\}$ and $M_g = \sup\{g(x) | x \in [a, b]\}$. Prove that

$$M_f - M_g < \varepsilon.$$

Additional Exercise 2. Let $|f(x) - g(x)| \leq \frac{\varepsilon}{2}$ for all $x \in [a, b]$. Let $M_f = \sup\{f(x) | x \in [a, b]\}$ and $M_g = \sup\{g(x) | x \in [a, b]\}$. We want to show that $M_f - M_g < \varepsilon$.

Choose $\delta = \frac{\varepsilon}{2}$. Now, consider M_f . We have that there exists $x_0 \in [a, b]$ such that $f(x_0) = M_f$. Therefore, we have $M_f - \delta = f(x_0) - \delta$.

Consider the expression

$$M_g \geq g(x_0) \geq f(x_0) - \frac{\epsilon}{2} > M_f - \delta - \frac{\epsilon}{2}$$

.

We can subtract $g(x_0)$ to get:

$$M_g - g(x_0) \geq g(x_0) - g(x_0) \geq f(x_0) - g(x_0) - \frac{\epsilon}{2} > M_f - \delta - \frac{\epsilon}{2} - g(x_0)$$

$$M_g - g(x_0) \geq 0 \geq f(x_0) - g(x_0) - \frac{\epsilon}{2} > M_f - \delta - \frac{\epsilon}{2} - g(x_0)$$

We can ignore the 0 to get

$$M_g - g(x_0) \geq f(x_0) - g(x_0) - \frac{\epsilon}{2} > M_f - \delta - \frac{\epsilon}{2} - g(x_0)$$

Then, we get add $g(x_0)$ to get .

$$M_g \geq f(x_0) - \frac{\epsilon}{2} > M_f - \delta - \frac{\epsilon}{2}$$

As $f(x_0) = M_f$, we can substitute to get

$$M_g - M_f \geq -\frac{\epsilon}{2} > -\delta - \frac{\epsilon}{2}$$

$$M_f - M_g \leq \frac{\epsilon}{2} < \delta + \frac{\epsilon}{2}$$

As $\delta = \frac{\epsilon}{2}$

$$M_f - M_g \leq \frac{\epsilon}{2} < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon$$

Hence, we have that $M_f - M_g < \epsilon$. □

3. a) Let $f: [a, b] \rightarrow \mathbb{R}$ and $M \in \mathbb{R}$. Show that if f is differentiable on (a, b) with $|f'(x)| \leq M, \forall x \in (a, b)$, then f is uniformly continuous on (a, b) .
- b) Give an example to show that the converse of a) fails.
- c) Prove that if $f: [0, \infty) \rightarrow \mathbb{R}$ is differentiable and f' is increasing and unbounded on $[0, \infty)$, then f is not uniformly continuous on $[0, \infty)$.

Additional Exercise 3(a.) Let $f: [a, b] \rightarrow \mathbb{R}$ and $M \in \mathbb{R}$. Suppose that f is differentiable on (a, b) such that $|f'(x)| \leq M$ for all $x \in (a, b)$. We want to show that f is uniformly continuous on (a, b) .

Let $\epsilon > 0$ and set $\delta = \frac{\epsilon}{M}$. Let $x, y \in (a, b)$ such that $|x - y| < \delta$. By **Corollary 12.15**, there exists a point $c \in (x, y)$ such that $f(y) - f(x) = f'(c)(y - x)$. As we have that $|f'(c)| \leq M$, we have that:

$$|f(y) - f(x)| = |f'(c)||y - x| \leq M|y - x| < M\delta = \epsilon.$$

Since δ depends only on ϵ and M , and not on the particular choice of x or y , this holds for all such pairs, and hence f is uniformly continuous on (a, b) . $\square$

Additional Exercise 3(b). It suffices to show that there exists a function that is uniformly continuous but not differentiable.

Define $f : [1, 2] \rightarrow \mathbb{R}$ as

$$f(x) = \begin{cases} x, & 1 \leq x \leq 1.5, \\ 3 - x, & 1.5 < x \leq 2. \end{cases}$$

f is continuous on a compact domain, hence it is uniformly continuous.

We note that $\lim_{h \rightarrow 0^-} \frac{f(1.5+h) - f(1.5)}{h} = \frac{(1.5+h) - 1.5}{h} = \frac{h}{h} = 1$ while $\lim_{h \rightarrow 0^+} \frac{f(1.5+h) - f(1.5)}{h} = \frac{(3 - (1.5+h)) - (3 - 1.5)}{h} = \frac{1.5 - h - 1.5}{h} = \frac{-h}{h} = -1$.

Hence, f is not differentiable at $x = 1.5$.

This completes the proof. $\square$

Additional Exercise 3(c.) Suppose for sake of contradiction that f is uniformly continuous on $[0, \infty)$. Let $\epsilon = 1$, then there exists $\delta > 0$ such that for all $x, y \in [0, \infty)$, if $|x - y| < \delta$ then $|f(x) - f(y)| < 1$.

Since f' is unbounded on $[0, \infty)$, there exists $z \in [0, \infty)$ such that $f'(z) > \frac{3}{\delta}$. Let $x = z$ and $y = z + \frac{\delta}{2}$, so that $|x - y| = \frac{\delta}{2} < \delta$.

By the Mean Value Theorem, there exists $m \in (x, y)$ such that:

$$f(y) - f(x) = f'(m)(y - x).$$

Since $m > x = z$ and f' is increasing, we have $f'(m) \geq f'(z) > \frac{3}{\delta}$. Therefore:

$$|f(y) - f(x)| = f'(m)(y - x) > \frac{3}{\delta} \cdot \frac{\delta}{2} = \frac{3}{2} > 1.$$

This contradicts $|f(x) - f(y)| < 1$, and hence f is not uniformly continuous on $[0, \infty)$. $\square$

4. a) Let $x_0 \in [a, b]$ be fixed and $f : [a, b] \rightarrow \mathbb{R}$ and $g : [a, b] \rightarrow \mathbb{R}$ be bounded functions such that $f(x) = g(x)$, $\forall x \neq x_0$. Prove that if g is integrable then f is also integrable and $\int_a^b f = \int_a^b g$.
- b) Let $f : [a, b] \rightarrow \mathbb{R}$ and $g : [a, b] \rightarrow \mathbb{R}$ be bounded functions such that $f(x) = g(x)$ for all but finitely many $x \in [a, b]$. Prove that if g is integrable then f is also integrable and $\int_a^b f = \int_a^b g$.
- c) Give an example of an integrable function $g : [a, b] \rightarrow \mathbb{R}$ and a function $f : [a, b] \rightarrow \mathbb{R}$ such that $g(x) = f(x)$ for all but countably many $x \in [a, b]$ and f is not integrable on $[a, b]$.
- d) Show that a bounded function $f : [a, b] \rightarrow \mathbb{R}$ that has only finitely many points of discontinuity is integrable. Is it necessary to have only finitely many discontinuities?

Additional Exercise 4(a.) Let g and f be bounded functions such that $f(x) = g(x)$ for all $x \neq x_0$. We want to show that if g is integrable then we have that f is integrable and that $\int_a^b f = \int_a^b g$.

We can define a function $h(x) = g(x) - f(x)$. By definition, we have that $h(x) = 0$ for all $x \neq x_0$. Then, we can define the function as

$$h(x) = \begin{cases} c & x = x_0 \\ 0 & x \neq x_0 \end{cases}$$

where $c = g(x_0) - f(x_0)$.

We first want to show that h is integrable with $\int_a^b h = 0$.

Since $h(x) = 0$ for all $x \neq x_0$, every subinterval of any partition contains points where $h = 0$, so $m_i(h) = 0$ for every subinterval. Therefore $L(h, P) = 0$ for every partition P , and hence $L(h) = 0$.

For the upper integral, let $\epsilon > 0$. Take a partition P that isolates x_0 in a subinterval of width $\delta < \frac{\epsilon}{|c|}$. Then $U(h, P) \leq |c|\delta < \epsilon$. Since ϵ was arbitrary, $U(h) = 0$.

Since $L(h) = U(h) = 0$, h is integrable and $\int_a^b h = 0$ by **Definition 13.16**.

Since $f = g - h$ and both g and h are integrable, f is integrable by **Theorem 13.25**. Then, we have that

$$\int_a^b f = \int_a^b g - \int_a^b h = \int_a^b g - 0 = \int_a^b g.$$

This suffices to complete the proof. □

Additional Exercise 4(b.) Let $f, g : [a, b] \rightarrow \mathbb{R}$ be bounded functions such that $f(x) = g(x)$ for all but finitely many $x \in [a, b]$. Let $X = \{x_1, x_2, \dots, x_n\}$ where X is a set of finitely many points where $f(x) \neq g(x)$. Then, we can define the function $h = g - f$, such that

$$h(x) = \begin{cases} g(x_i) - f(x_i) & x = x_i \text{ for some } i \in \{1, \dots, n\} \\ 0 & \text{otherwise.} \end{cases}$$

We first show that h is integrable with $\int_a^b h = 0$.

Since $h(x) = 0$ for all $x \notin \{x_1, \dots, x_n\}$, every subinterval of any partition contains points where $h = 0$, so $m_i(h) = 0$ for every subinterval. Therefore $L(h, P) = 0$ for every partition P , and hence $L(h) = 0$.

For the upper integral, let $M = \max_i |h(x_i)|$ and let $\epsilon > 0$. Take a partition P that isolates each x_i in its own subinterval of width $\delta < \frac{\epsilon}{nM}$. Then $U(h, P) \leq nM\delta < \epsilon$. Since ϵ was arbitrary, $U(h) = 0$.

Since $L(h) = U(h) = 0$, by **Definition 13.16**, we have that h is integrable and $\int_a^b h = 0$. Since $f = g - h$ and both g and h are integrable, by **Theorem 13.25** we have that f is integrable and

$$\int_a^b f = \int_a^b g - \int_a^b h = \int_a^b g - 0 = \int_a^b g.$$

This suffices to complete the proof. □

Additional Exercise 4(c.) Define $g : [0, 1] \rightarrow \mathbb{R}$ as $g(x) = 73$ and define $f : [0, 1] \rightarrow \mathbb{R}$ as

$$f(x) = \begin{cases} 21 & x \in \mathbb{Q} \\ g(x) & x \notin \mathbb{Q}. \end{cases}$$

We know from **Exercise 13.17** that g is integrable. As we have that $f(x) = g(x)$ for all $x \notin \mathbb{Q}$, and as $\mathbb{Q}$ is countable, we have that $f(x) = g(x)$ for all but countably many points. For any partition P of $[0, 1]$, every subinterval contains both rational and irrational points, so $L(f, P) = 21$ and $U(f, P) = 73$ for every partition P . Hence for $\epsilon = 7 > 0$, there does not exist a partition P such that $U(f, P) - L(f, P) < \epsilon$, so f is not integrable. □

Additional Exercise 4(d.) We proceed by induction on n , the number of points of discontinuity of f .

Suppose f is discontinuous at only one point $x_0 \in [a, b]$. Let $\epsilon > 0$.

Since f is bounded, there exists $M \in \mathbb{R}$ such that $|f| < M$. Then, consider the points $x_0 - \delta$ and $x_0 + \delta$, where $\delta > 0$ is to be chosen later. Since f is continuous on $[a, x_0 - \delta]$ and $[x_0 + \delta, b]$, by **Theorem 13.19**, we have that it is integrable on each. Thus, there exist partitions P_1 of $[a, x_0 - \delta]$ and P_2 of $[x_0 + \delta, b]$ such that

$$U(f, P_1) - L(f, P_1) < \frac{\epsilon}{3} \quad \text{and} \quad U(f, P_2) - L(f, P_2) < \frac{\epsilon}{3}.$$

Now, let $P = P_1 \cup P_2$. Then

$$U(f, P) < U(f, P_1) + 2M\delta + U(f, P_2) \quad \text{and} \quad L(f, P) > L(f, P_1) - 2M\delta + L(f, P_2).$$

Therefore

$$U(f, P) - L(f, P) < (U(f, P_1) - L(f, P_1)) + 4M\delta + (U(f, P_2) - L(f, P_2)) < \frac{2\epsilon}{3} + 4M\delta.$$

Then, we can choose $\delta = \frac{\epsilon}{12M}$ to then give us $U(f, P) - L(f, P) < \epsilon$. By **Theorem 13.18**, f is integrable on $[a, b]$. This works to complete the base case.

Now, suppose the result holds for any bounded function with at most n points of discontinuity. Let $f : [a, b] \rightarrow \mathbb{R}$ be bounded with $n + 1$ points of discontinuity, and let $x_0 \in [a, b]$ be one such point.

Then f restricted to $[a, x_0]$ and $[x_0, b]$ each have at most n points of discontinuity. By the inductive hypothesis, f is integrable on each. By **Theorem 13.23**, f is integrable on $[a, b]$. Thus, this suffices to complete the proof. $\square$

5. Suppose that $\varphi : \mathbb{R} \rightarrow \mathbb{R}$ is continuous and $f : \mathbb{R} \rightarrow \mathbb{R}$ is bounded and integrable on $[a, b]$. Show that $\varphi \circ f$ is integrable on $[a, b]$.